

Why are flights late?

A 50-million-flight study of U.S. on-time performance, 2018–2025.

WHY THIS MATTERS

~800M

U.S. Annual Passengers

A single 30-minute delay cascades into missed connections, idle crews, blocked gates, and angry customers.

Three operational decisions hinge on understanding delay behavior:

- **How big are capacity buffers?** Is the average delay a meaningful planning number?
- **Are some carriers really better?** Or is the dashboard's ranking just sampling noise?
- **Can we predict risk?** Can we forecast next year's delay environment at booking time?

THE DATA

Source: U.S. Bureau of Transportation Statistics — On-Time Performance reports.

49.7M

Total flights

4.79

Avg arrival delay (min)

81.3%

On-time arrival

2.22%

Cancellation rate

These four numbers anchor the whole project — they match the Power BI dashboard exactly.

But the average is hiding something important. Let's look at the distribution.

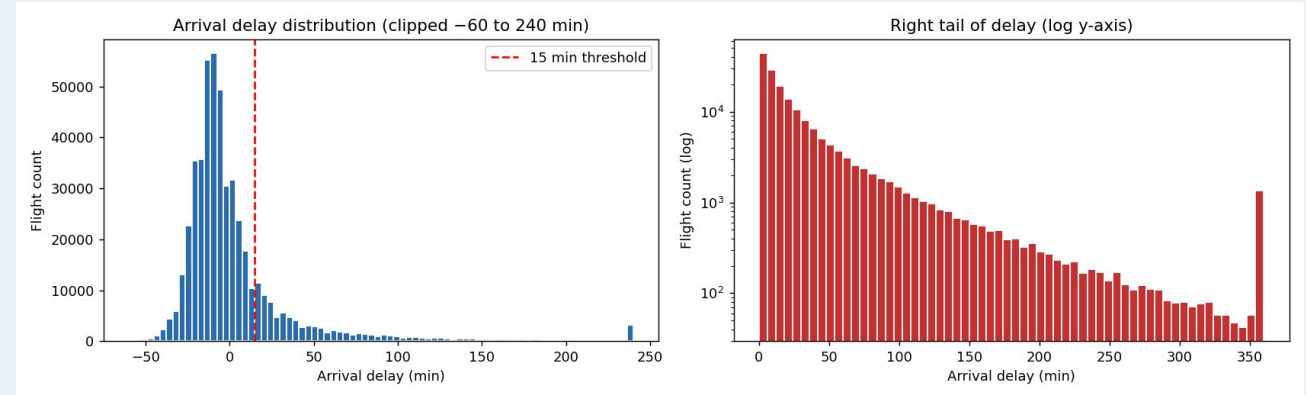
THE DELAY DISTRIBUTION

What this chart shows

Every U.S. flight, plotted by how late it arrived. Most cluster near zero — but a long tail stretches to the right.

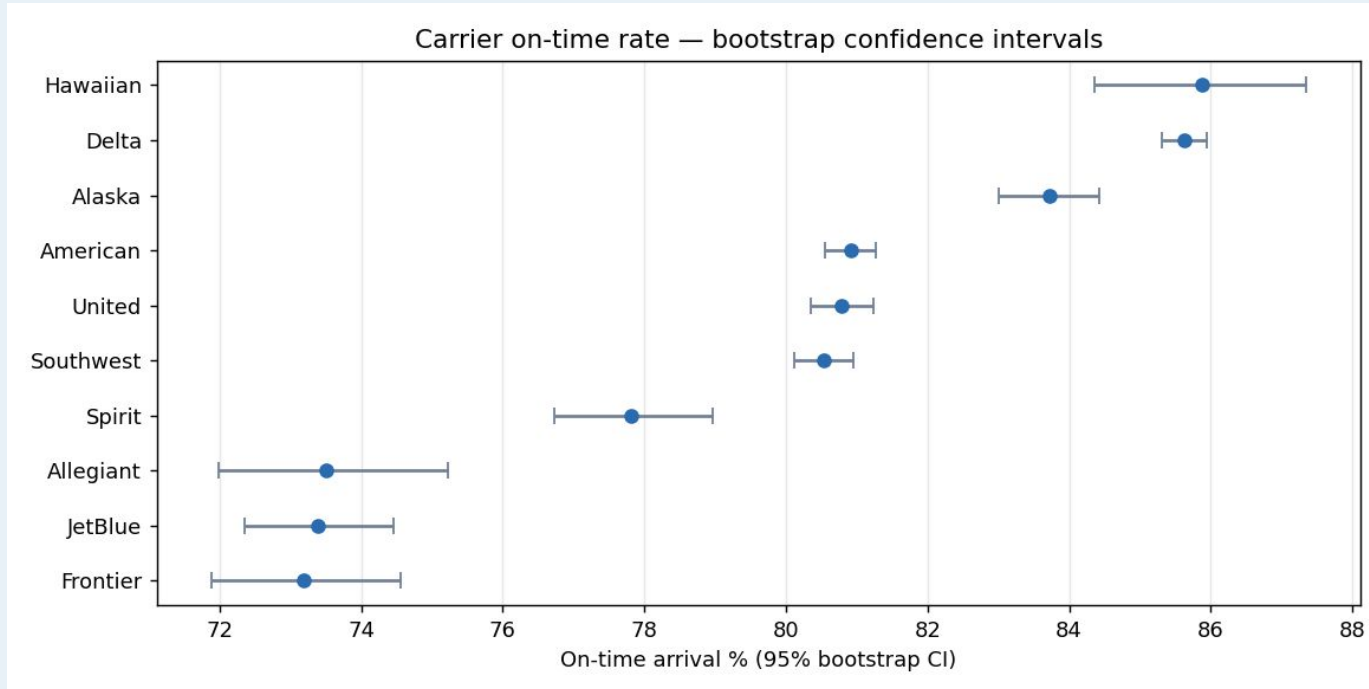
What that means

- ▶ Mean: 4.77 min | σ : 52.85 min (11× the mean!)
- ▶ 1 in 5 flights is at least 15 min late
- ▶ 1 in 16 is at least 1 hour late
- ▶ 1 in 84 is at least 3 hours late



KEY TAKEAWAY: Its better to against the tail rather than the mean here.

There a real difference between best and worst carriers.



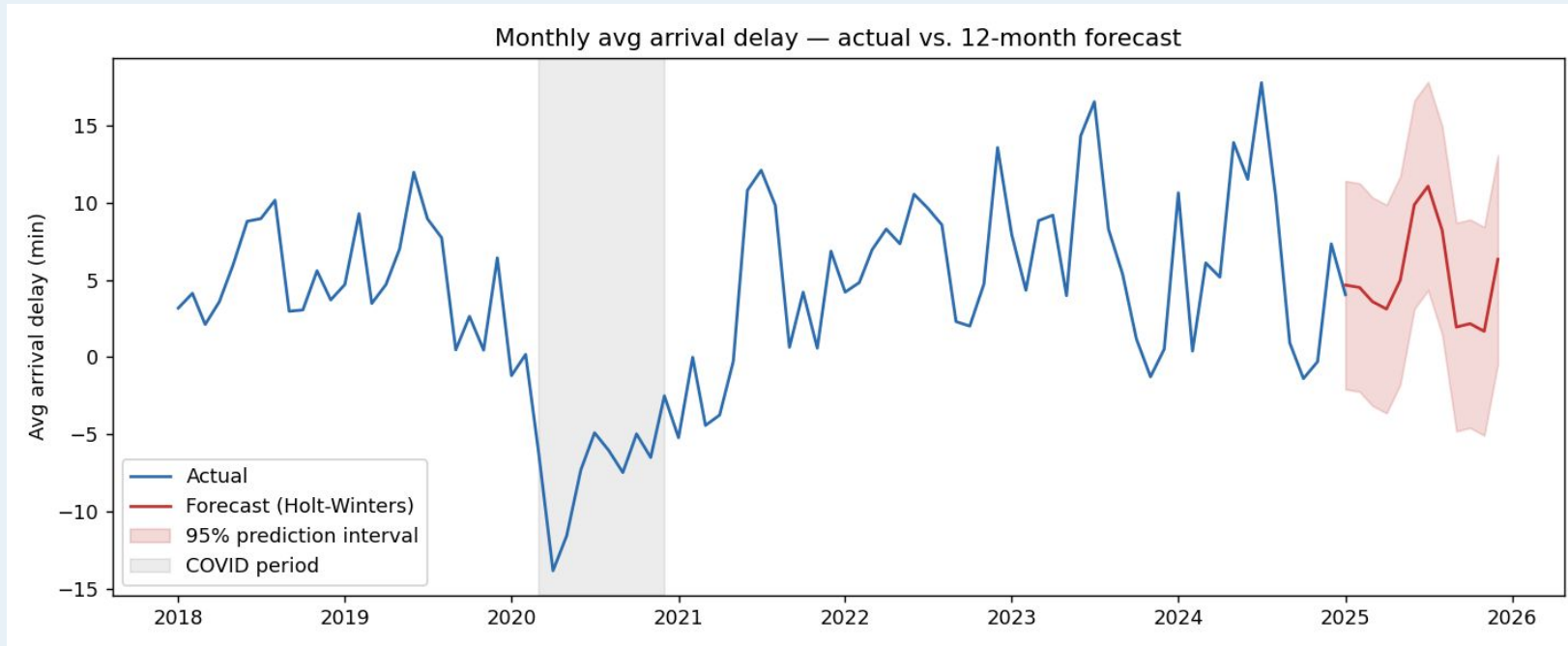
What this chart shows

Each carrier's on-time rate, with horizontal error bars showing the 95% confidence interval from a bootstrap of 200K flights.

Why CIs matter here

- ▶ Hawaiian / Delta lead at $\approx 86\%$
- ▶ Frontier / JetBlue trail at $\approx 73\%$
- ▶ The CIs at the top do NOT overlap with the CIs at the bottom
- ▶ → The ranking is a real operational difference

Yearly Arrival Delays



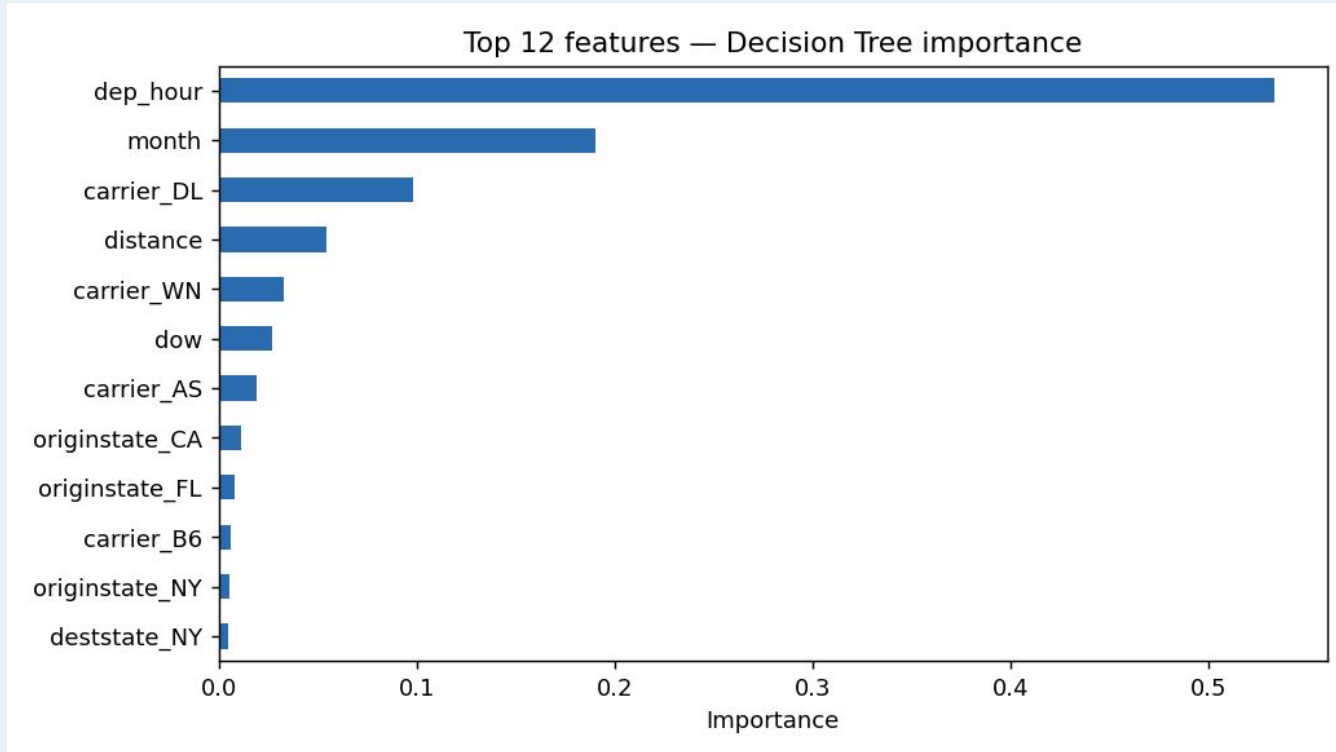
What we did

Holt-Winters exponential smoothing on 85 months of average delay. Train ends Dec 2024; the red curve is a 12-month forecast.

Two clear signals

- ▶ Stable annual seasonality: Jun–Aug peaks every year
- ▶ 2020 is a structural break — must be excluded from forward models

PREDICTING AT-RISK FLIGHTS



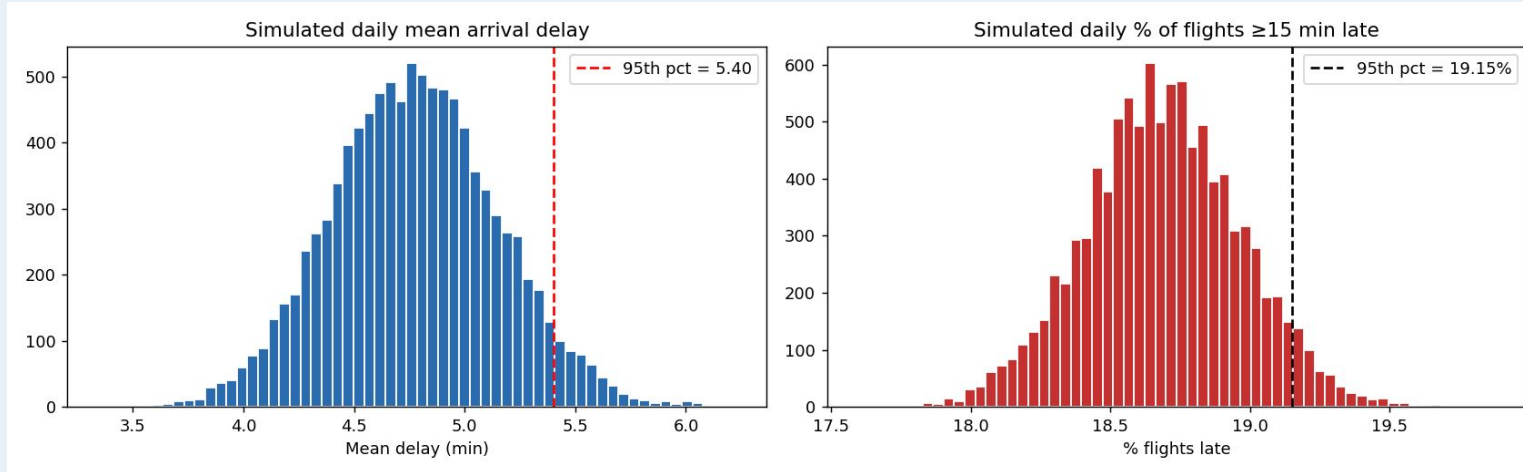
Model: depth-8 decision tree

Inputs (known at booking): carrier, month, day-of-week, scheduled departure hour, origin/destination state, distance.

Performance

- ▶ ROC-AUC: 0.626 (real signal)
- ▶ Recall: 60% — flags 6 of every 10 late flights
- ▶ Precision: 24.9% — when flagged, 1.3× base rate
- ▶ Top driver: scheduled departure hour

SIMULATING BAD DAYS



What we did

Simulated 10,000 hypothetical days × 20,000 flights/day, drawing from the empirical delay distribution.

Results

- ▶ Typical day: 18.69% late
- ▶ 95th-pct day: 19.15% late
- ▶ Worst sim'd day: 19.88% late

CAVEAT: Real bad days are correlated (weather, ATC) — the per-flight tail is where the pain lives.

WHAT TO DO ABOUT IT

1

Re-time evening departures

Pull marginal late-evening flights forward 30–60 min. Departure hour is the tree's #1 feature, and late-evening flights seed next-day cascading delays.

2

Staff a +25% summer buffer

Holt-Winters seasonality + heavy-tail probability both flag June–August. Don't size to the annual mean — size to the 95th-pct summer day.

3

Price contracts to CIs

Carrier on-time gaps are statistically real. Code-share economics, SLAs, and corporate-travel contracts should reflect each carrier's confidence interval.

Honest limits of this analysis.

Data Scope

BTS only includes carriers $\geq 0.5\%$ domestic share.
Small regionals are absent.

Missing Predictors

No weather, ATC flow, or aircraft-position data—known dominant predictors are missing, capping accuracy.

Modeling Choice

Decision tree treats delay as binary. A delay-minutes regression would prioritize better.

Independence Assumption

Monte Carlo assumes independent delays. Real bad days are correlated, so true tail risk is wider.

Seasonal Stability

Holt-Winters assumes stable post-COVID patterns.
New shocks (fuel, labor) would invalidate forecasts.

Uncontrolled Route Mix

Carrier ranking doesn't control for route mix—e.g., Hawaiian flies short, uncongested routes.

Flight Delay Analysis BTS

Overview

By Cause

By Carrier

Geography

Day x Month Heatmap

50M

Total Flights

4.79

Avg Arrival Delay

81.3%

On-Time Arrival

2.2%

Cancellation %

Delay Definition

Arr +15 Min
Late

Dep/Arr +15
Min Late

Time Grain

Week

Month

Quarter

Year

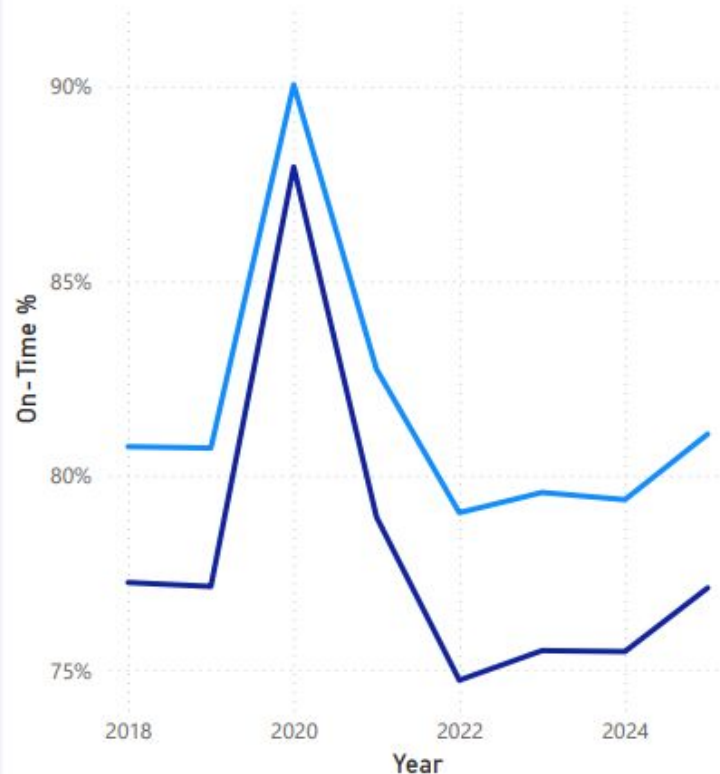
Date

1/1/2018

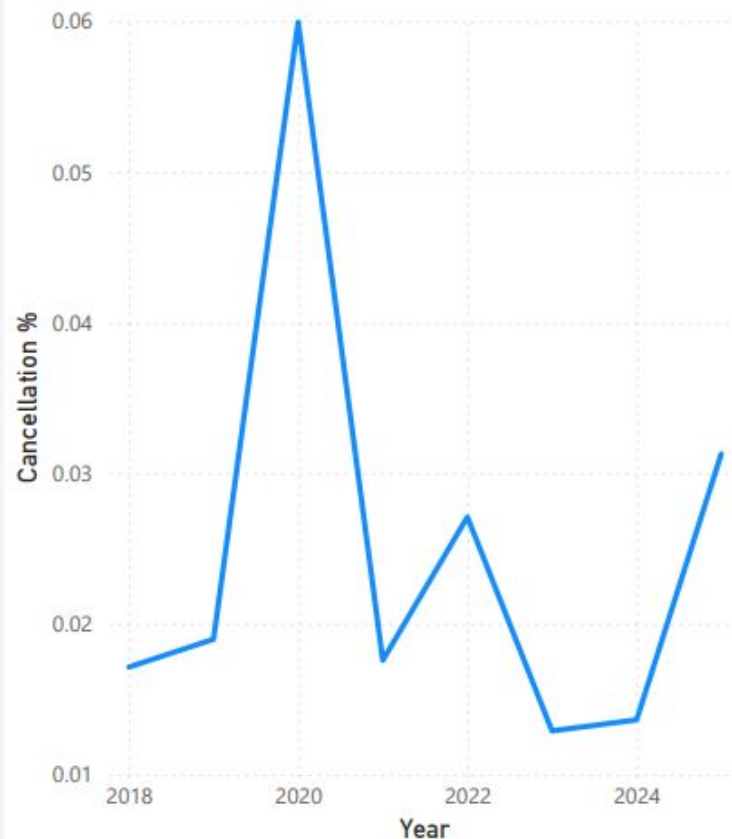
1/31/2025

On-Time Performance Over Time

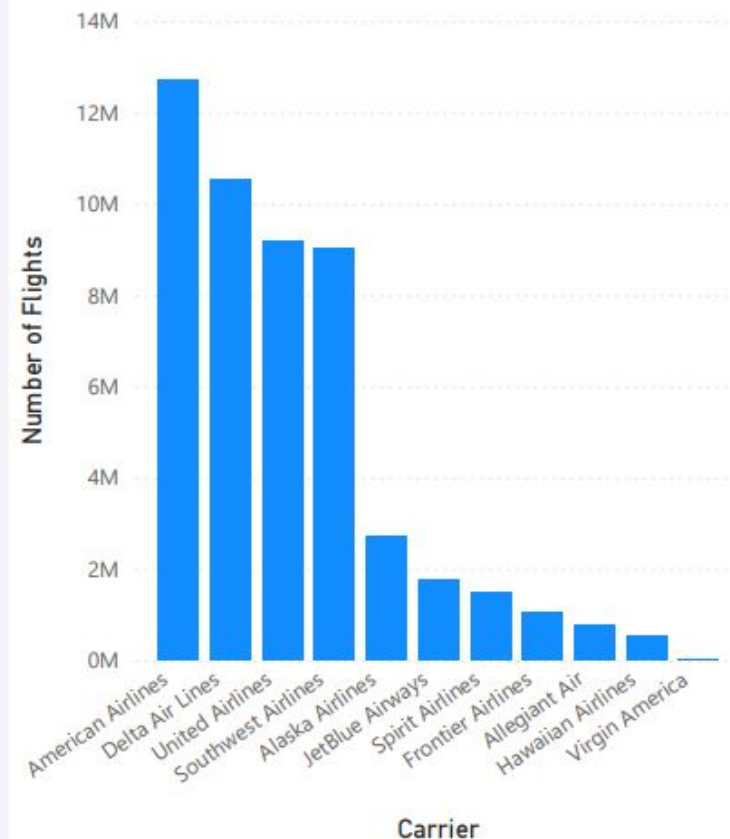
● Arr +15 Min Late ● Dep/Arr +15 Min Late



Cancellation % by Year



Total Flights by Carrier



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50M

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Cancellation %

Delay Definition

Arr +15 Min
Late

Dep/Arr +15
Min Late

Time Grain

Week

Month

Quarter

Year

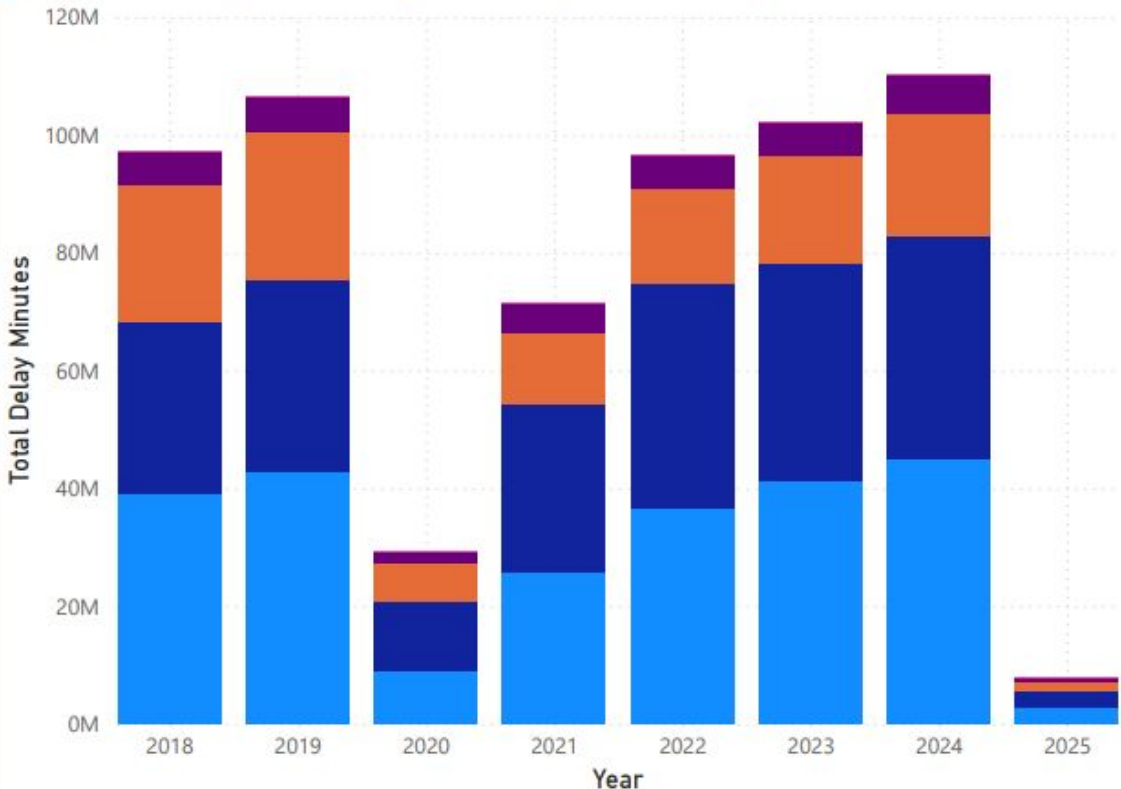
Date

1/1/2018

1/31/2025

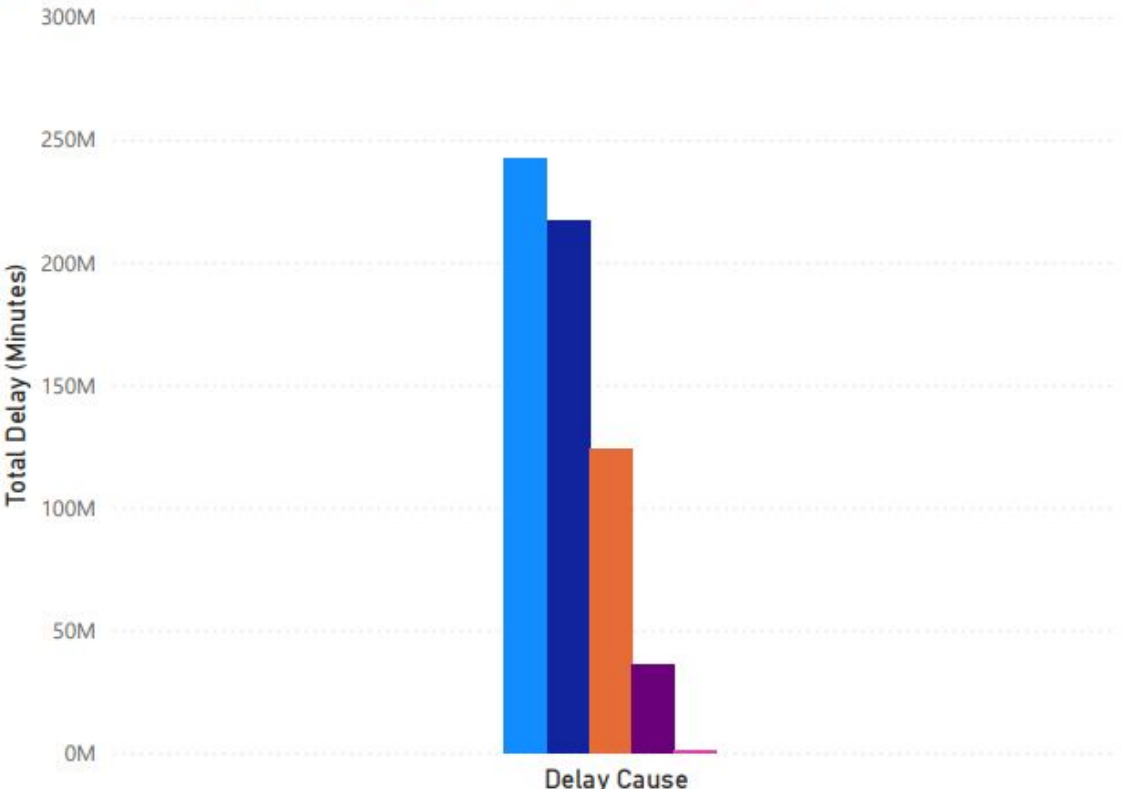
Delay Breakdown by Cause

Late Aircraft Delay Carrier Delay NAS Delay Weather Delay Security Delay



Total Delay by Cause

Late Aircraft Delay Carrier Delay NAS Delay Weather Delay Security Delay



Flight Delay Analysis BTS

[Overview](#)[By Cause](#)[By Carrier](#)[Geography](#)[Day x Month Heatmap](#)**15M**

Total Flights

5.58

Avg Arrival Delay

80.7%

On-Time Arrival

1.9%

Cancellation %

Delay Definition

Arr +15 Min
LateDep/Arr +15
Min Late

Time Grain

Week

Month

Quarter

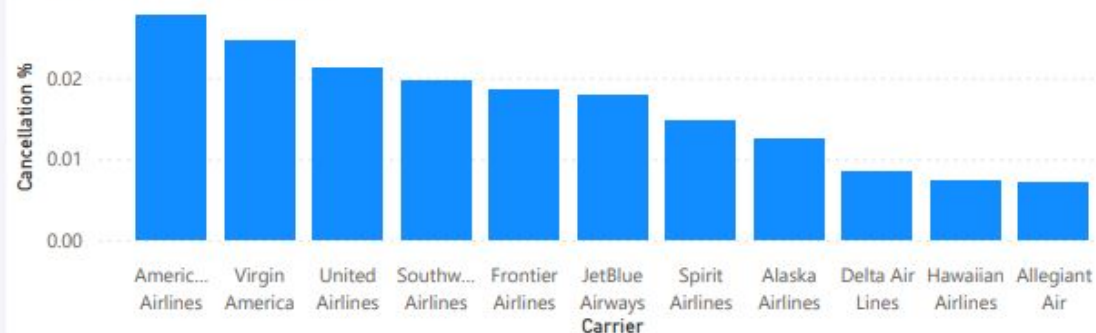
Year

Date

1/1/2018

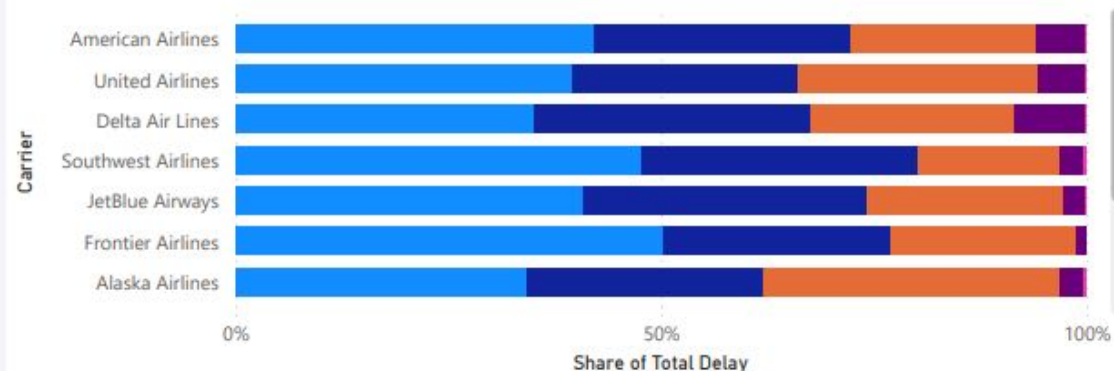
11/13/2019

Cancellations by Carrier

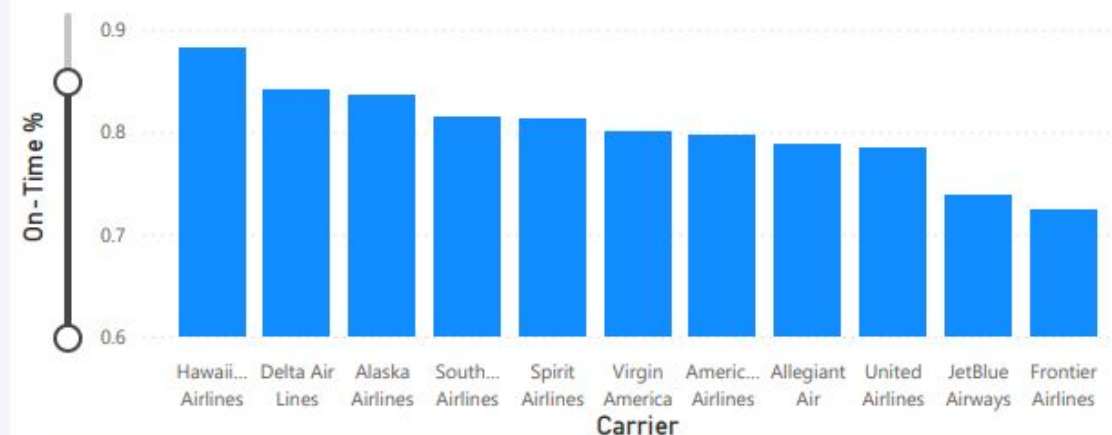


Delay Cause Mix by Carrier

Late Aircraft Delay Carrier Delay NAS Delay Weather Delay Security Delay



On-Time % By Carrier



Total Flights by Carrier



Flight Delay Analysis BTS

Overview

By Cause

By Carrier

Geography

Day x Month Heatmap

50M

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Cancellation %

Geography Dimension

Origin Airport

Origin City

Destination Airport

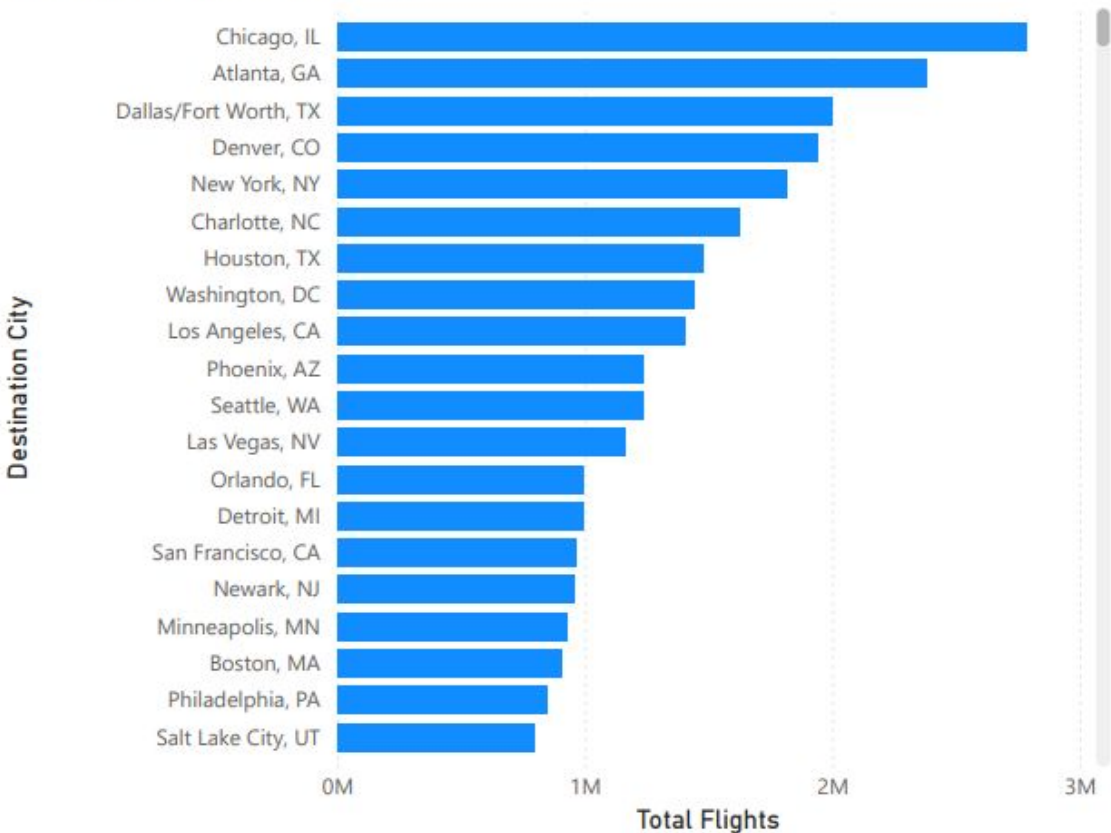
Destination City

Date

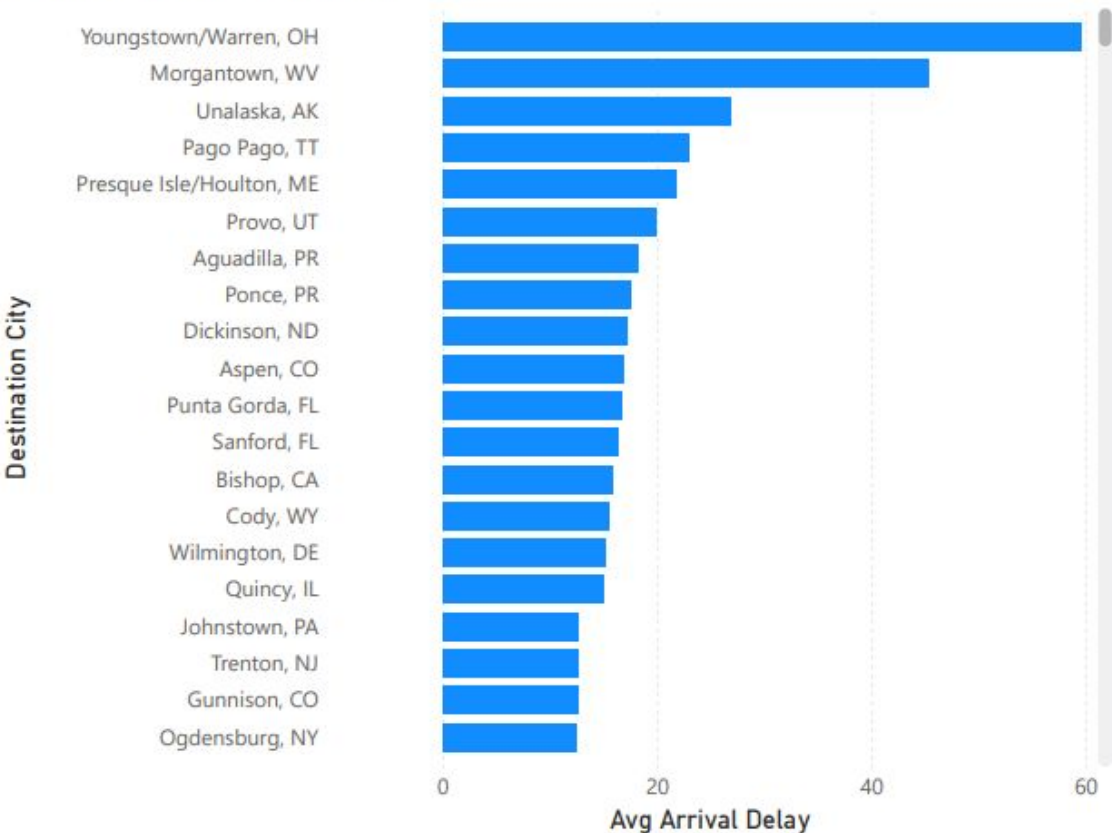
1/1/2018

1/31/2025

Total Flights by Destination City



Avg Arrival Delay by Destination City



Flight Delay Analysis BTS

[Overview](#)[By Cause](#)[By Carrier](#)[Geography](#)[Day x Month Heatmap](#)**50M**

Total Flights

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Avg Arrival Delay

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Cancellation %

Delay Definition

Arr +15 Min
LateDep/Arr +15
Min Late

Time Grain

Week

Month

Quarter

Year

Date

1/1/2018

1/31/2025

Total Flights by Day of Week & Month

Day Name	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Mon	703032	555757	622533	611634	589333	568771	669261	633807	609634	633544	603473	613881
Tue	651402	513353	601587	547053	556746	555697	616672	600705	528061	606151	567721	561167
Wed	683230	522065	609030	529100	591455	575686	611531	613541	551963	594605	579611	555545
Thu	692762	579463	650621	574166	601863	601852	604523	673661	608797	629160	555961	607742
Fri	683738	561480	674984	585631	589571	606521	630702	652259	609403	611221	612644	611877
Sat	546872	473589	562540	499369	466992	552158	575901	555968	504384	523268	540945	540045
Sun	658000	526335	637707	572388	551559	594274	632131	605784	606064	622166	581953	633532

Cancellation % by Day of Week & Month

Day Name	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Mon	3.3%	2.2%	4.4%	4.1%	1.6%	2.1%	2.2%	2.5%	0.9%	0.8%	1.0%	1.8%
Tue	3.0%	2.2%	4.6%	3.7%	1.5%	1.8%	1.9%	2.3%	1.1%	0.8%	0.6%	1.5%
Wed	2.7%	3.3%	4.2%	5.6%	1.5%	1.9%	1.6%	2.1%	1.3%	0.9%	0.6%	1.7%
Thu	2.8%	3.5%	3.8%	5.4%	1.5%	2.0%	1.8%	1.7%	1.8%	1.3%	0.9%	1.9%
Fri	3.0%	2.4%	3.9%	4.8%	1.9%	1.5%	2.2%	1.9%	1.8%	0.8%	0.9%	1.9%
Sat	3.1%	1.3%	3.9%	5.3%	1.4%	1.6%	2.0%	1.4%	1.1%	0.9%	0.5%	1.5%
Sun	3.2%	2.0%	3.9%	5.0%	1.5%	2.3%	2.3%	2.4%	0.9%	1.0%	0.5%	1.7%

Avg Arrival Delay by Day of Week & Month

Day Name	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Mon	6.38	2.58	1.60	4.49	6.30	10.48	12.17	8.46	1.66	2.04	1.85	5.69
Tue	2.89	0.03	-1.80	0.71	4.28	8.16	8.98	6.09	-0.74	-1.10	0.33	3.24
Wed	2.37	2.66	0.22	2.43	4.53	8.60	8.31	5.93	-0.35	-0.98	-0.84	3.30
Thu	2.07	6.59	4.34	4.04	9.09	11.43	10.20	9.89	3.38	3.35	1.42	6.00
Fri	4.33	6.35	5.56	5.90	7.04	10.69	12.98	9.71	4.62	3.79	2.31	6.71
Sat	3.16	1.32	4.56	3.90	2.45	9.34	10.33	5.03	-1.75	-0.98	-0.32	5.94
Sun	5.40	3.94	3.97	4.90	5.52	12.59	12.37	8.60	1.24	2.58	0.96	6.74

Thank you.
